

From Black to White Boxes – Interpretable Regression with the TRUST™ Algorithm

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Whitebox Lab

October 20, 2025

- Ph.D. in Statistics, University of Wisconsin-Madison (La Caixa Fellow)
- Former Risk Analyst, European Central Bank
 - Led early applications of predictive ML in my division (2017)
 - I probably wouldn't be here today without that experience!
- Began my coding journey at Carnegie Mellon University (2015) and have contributed 3 open-source R packages with >100K downloads
- Creator of the **TRUST™** algorithm and **trust-free** Python package

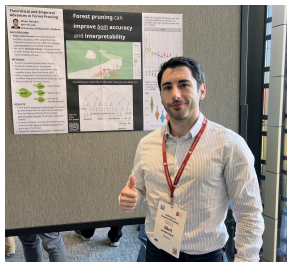


Figure 1: Presenting my forest-pruning paper at CPAL 2025 (Stanford University)

1. Review of Regression

- Main goal: to **estimate with enough accuracy** a **real-valued** outcome.
 - Examples: estimating **medical insurance charges** based on individual characteristics (later today), estimating a **disease marker** based on e.g. lifestyle choices, estimating **house prices** based on their attributes, estimating **demand for a good/service** based on certain variables
 - Formally: assuming $y = f(X) + \varepsilon$ (typically $\varepsilon \sim N(0, \sigma)$) we use data (X, y) to learn f from a set S of candidate models
- Other possible goals: **transparency**, fairness, efficiency
- Focus of this presentation: **accuracy + transparency = TRUST** (pun completely intended)
 - But **is that combination realistic** in practice? Not really – let's wrap up.

Just kidding. . . Although that was long-believed to be the case:

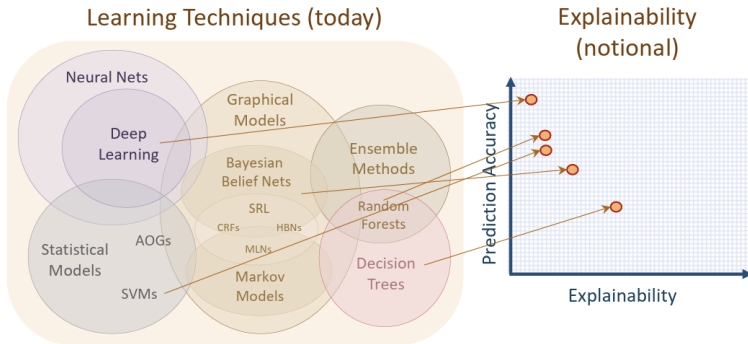


Figure 2: Source: DARPA's XAI Program (2016)

The reason for **hope**: multiple models (including, in some cases, *simple* ones) can describe the $X \rightarrow y$ map reasonably well (Rashomon Effect).

2. Interpretable Machine Learning Models

They include, in rough order of interpretability*:

- (Short) Decision Trees
- (Sparse) Linear Models and Generalized Linear Models (GLM's)
- (Sparse) Linear Model Trees
- (Sparse) Generalized Additive Models (GAM's) and friends
(e.g. Splines, RuleFit, NodeHarvest)

*Context-dependent and global vs local notions

3. Introducing TRUST™ (PRICAI 2025)

- **T**ransparent: global and local explanations
- **R**obust: smart handling of missing values and extra/interpolation
- **U**ltra-**S**parsity: relaxed Lasso leaves
- **T**rees: subgroup identification through axis-aligned partitions

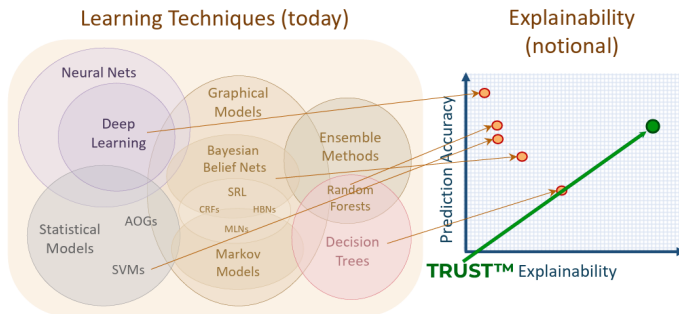


Figure 3: *Updated* DARPA's explainability landscape

4. Example: Modeling Medical Insurance Charges

Curious to see **TRUST™** in action on a real-world dataset? Check out our [Jupyter Notebook](#).